



Technical University of Denmark

DTU Compute

Department of Applied Mathematics and Computer Science

Query Optimization

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Query optimization

- Queries consist of several atomic operations
 - Selection, projection, join, etc....
- Operations can be executed in different order, as long as the result remains the same
 - E.g., selection then join vs join then selection
- The order in which operations must be performed is called execution plan
 - Although leading to the same result, execution plans can have substantial differences in processing time and memory usage
- Optimization finds the best execution plan
 - Completely carried out by the DBMS
 - Users do not have to optimize their queries

From SQL to relational algebra

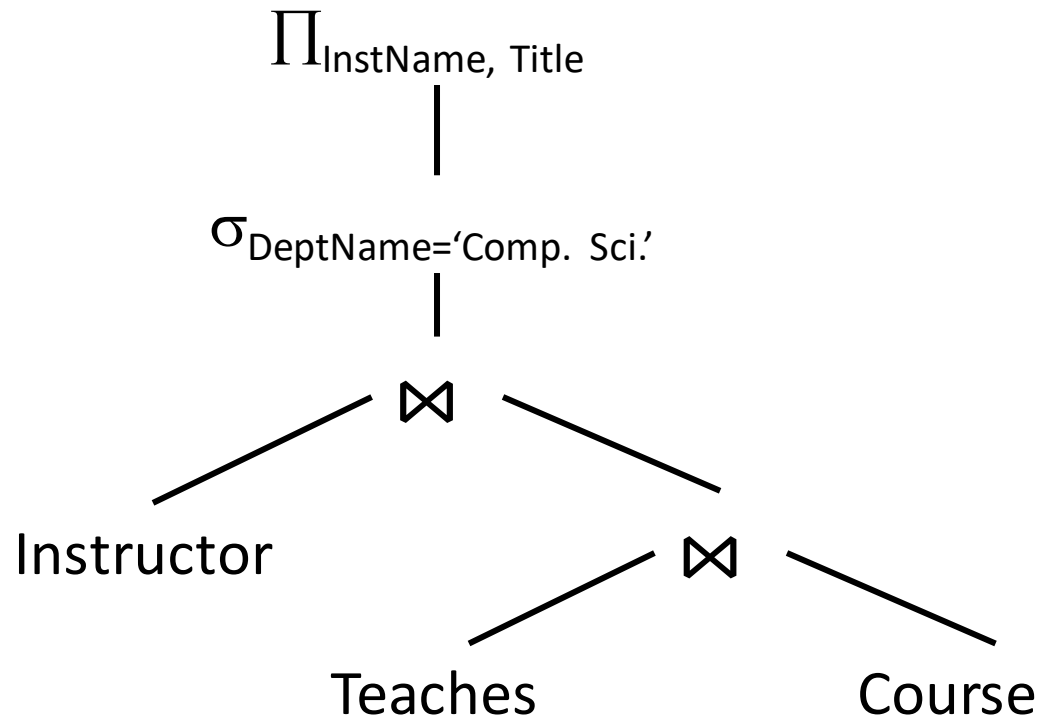
- Find the names of instructors in the Comp. Sci. department together with the course titles of the courses that the instructors teach.
 - `SELECT InstName, Title FROM Instructor NATURAL JOIN Teaches NATURAL JOIN Course WHERE deptname = 'Comp. Sci.'`
- This SQL query can be rewritten in relational algebra:
 - $\Pi_{\text{InstName, Title}} (\sigma_{\text{DeptName}='Comp. Sci.'} (\text{Instructor} \bowtie \text{Teaches} \bowtie \text{Course}))$
- From this expression, the DBMS applies transformation rules to calculate equivalent expressions.
- The DBMS then selects the one that minimizes the processing time
 - Some transformation rules always produce more efficient expressions
 - In the other cases, the DBMS estimates the cost of each operation, and selects the expression that minimizes the total cost.

Syntax tree

- Relational algebra expressions can be represented as a syntax tree:
 - Leaf nodes represent the relations
 - All other nodes represent the operations
 - The closer a node is from the leaves, the sooner the corresponding operation will be performed

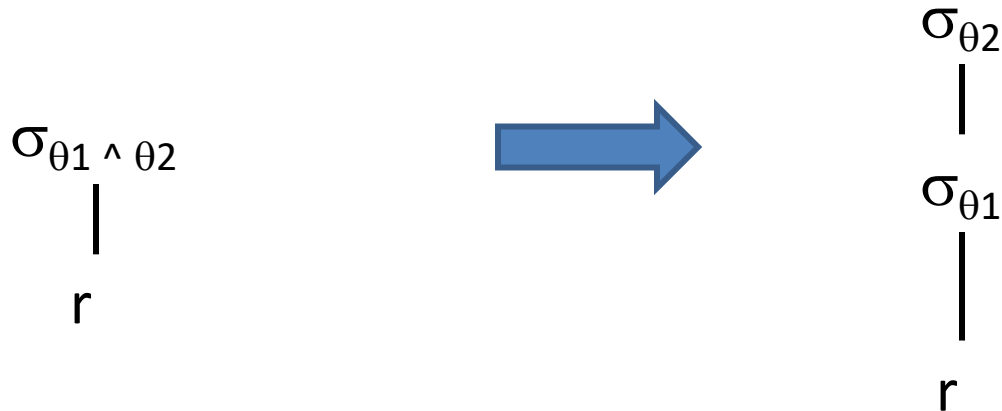
Syntax tree

- Find the names of instructors in the Comp. Sci. department together with the course titles of the courses that the instructors teach.
 - $\Pi_{\text{InstName, Title}} (\sigma_{\text{DeptName}='Comp. Sci.'} (\text{Instructor} \bowtie \text{Teaches} \bowtie \text{Course}))$



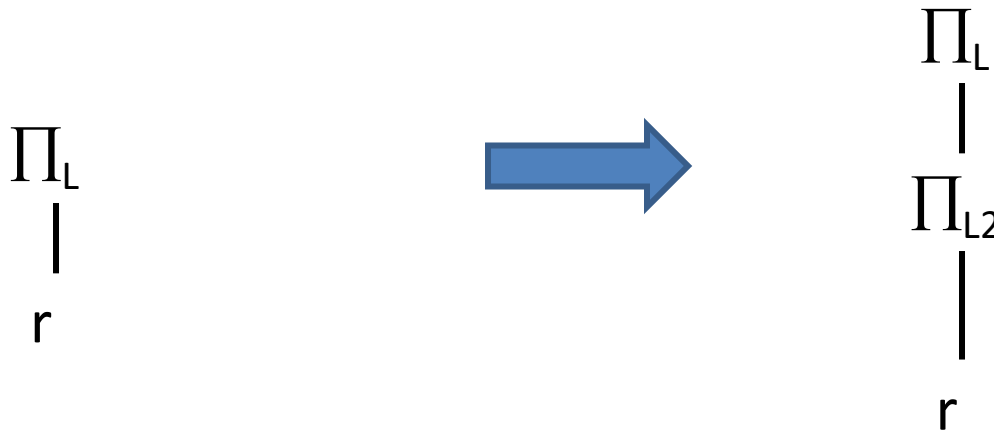
Equivalence of expressions

- Making selection operations atomic
 - Can be done only if θ_1 and θ_2 are in conjunction (^).



Equivalence of expressions

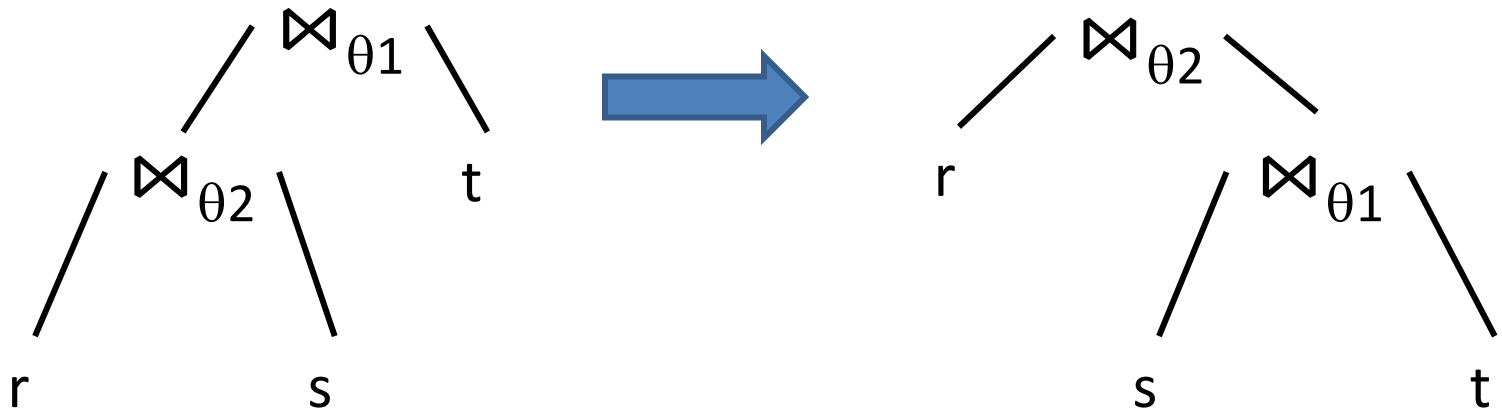
- Expanding projection operations
 - Can be done only if $L \subseteq L_2$.



Equivalence of expressions

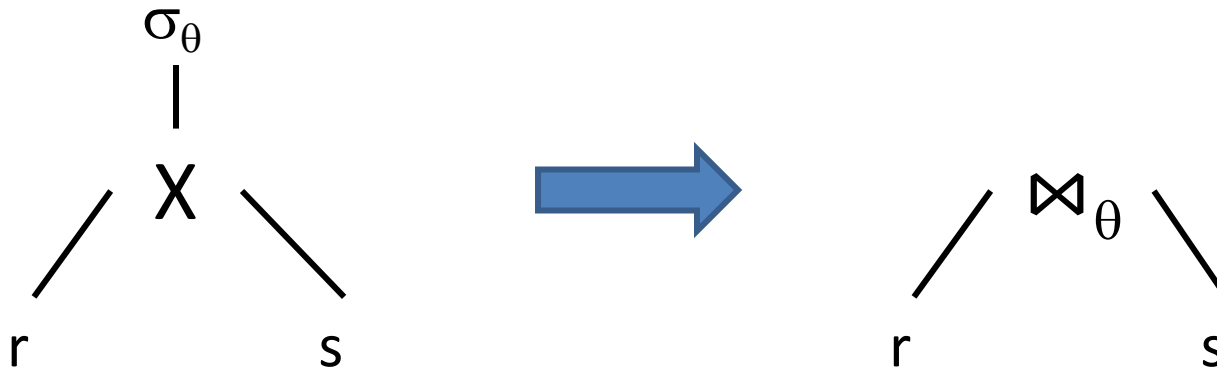
■ Switching natural joins

- Can be done only if θ_1 does **not** refer to attributes of r



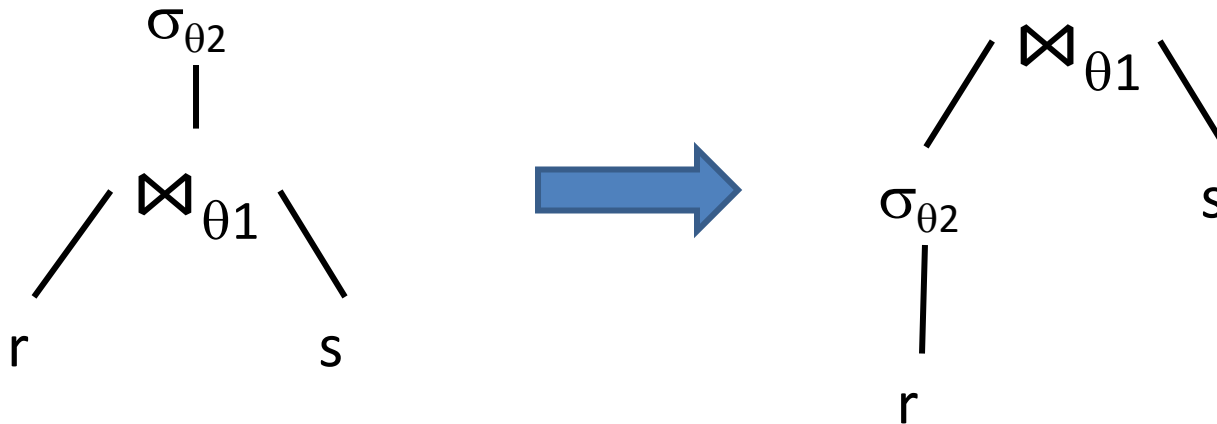
Query optimizations

- Removing cartesian product operations



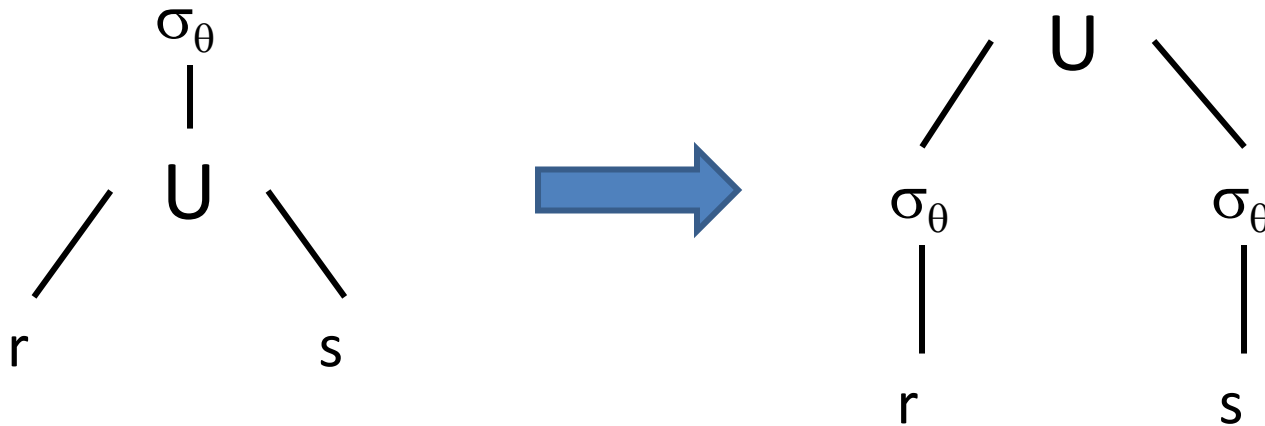
Query optimizations

- Pushing selection operations before joins
 - Can be done only if θ_2 refers only to attributes of r .



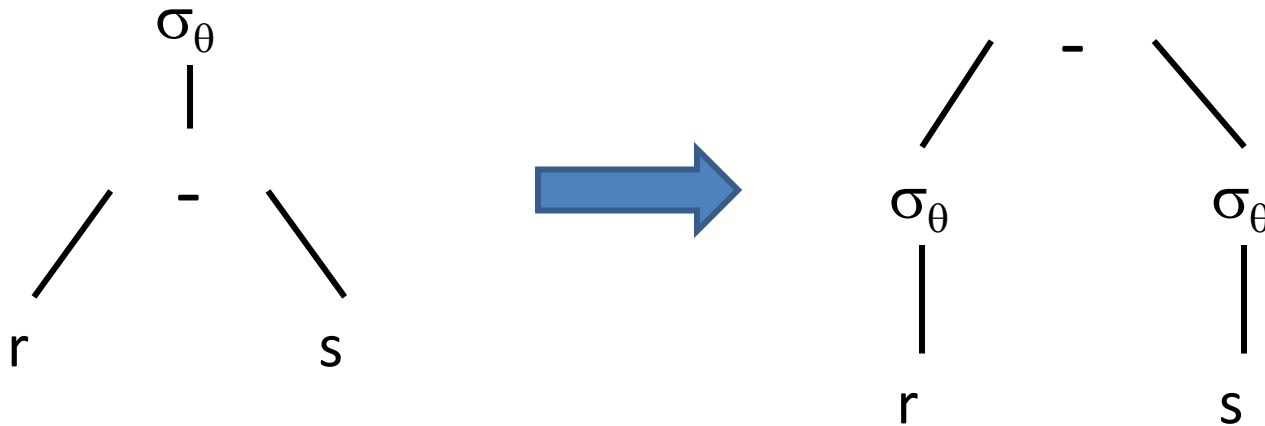
Query optimizations

- Pushing selection operations before unions



Query optimizations

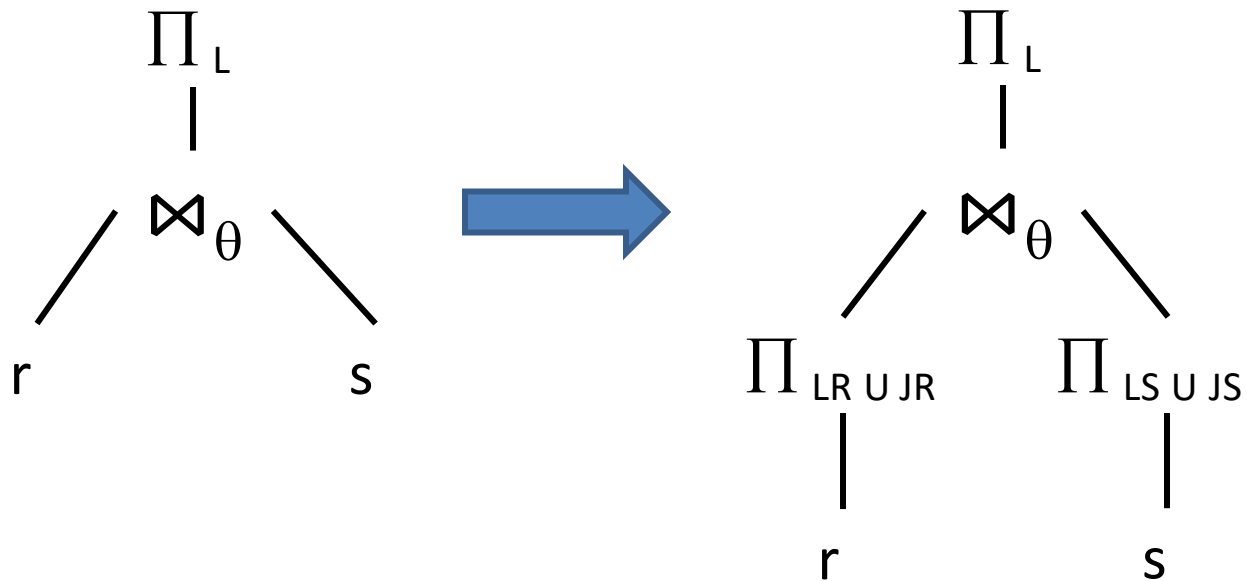
- Pushing selection operations before differences



Query optimizations

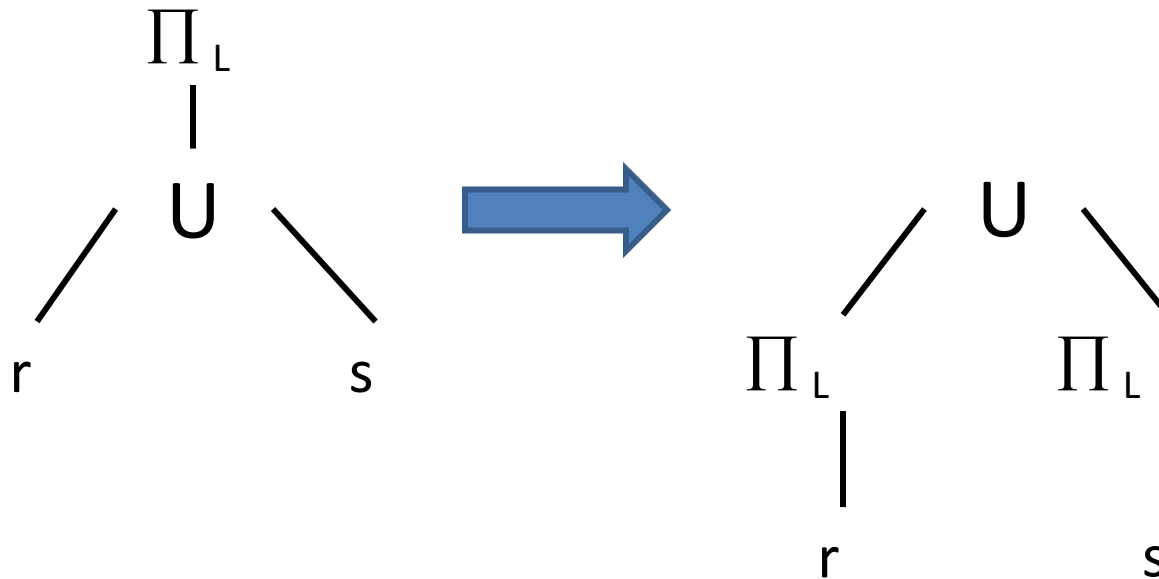
■ Pushing projection operations before joins

- $LR = L \cap \text{Schema}(r)$ $JR = \text{Attributes}(\theta) \cap \text{Schema}(r)$
- $LS = L \cap \text{Schema}(s)$ $JS = \text{Attributes}(\theta) \cap \text{Schema}(s)$



Query optimizations

- Pushing projection operations before unions



Back to the example

- Find the names of instructors in the Comp. Sci. department together with the course titles of the courses that the instructors teach.

InstName	Title
Srinivasan	Intro. to Computer Science
Srinivasan	Robotics
Srinivasan	Database System Concepts
Katz	Intro. to Computer Science
Katz	Image Processing
Brandt	Game Design
Brandt	Game Design
Brandt	Image Processing

$\Pi_{InstName, Title}$

$\sigma_{DeptName='Comp. Sci.'}$

InstID	DeptName	CourseID	SectionID	Semester	StudyYear	Title	Credits	InstName	Salary
10101	Comp. Sci.	CS-101	1	Fall	2009	Intro. to Computer Science	4	Srinivasan	65000.00
10101	Comp. Sci.	CS-315	1	Spring	2010	Robotics	3	Srinivasan	65000.00
10101	Comp. Sci.	CS-347	1	Fall	2009	Database System Concepts	3	Srinivasan	65000.00
45565	Comp. Sci.	CS-101	1	Spring	2010	Intro. to Computer Science	4	Katz	75000.00
45565	Comp. Sci.	CS-319	1	Spring	2010	Image Processing	3	Katz	75000.00
83821	Comp. Sci.	CS-190	1	Spring	2009	Game Design	4	Brandt	92000.00
83821	Comp. Sci.	CS-190	2	Spring	2009	Game Design	4	Brandt	92000.00
83821	Comp. Sci.	CS-319	2	Spring	2010	Image Processing	3	Brandt	92000.00



Instructor

Teaches

Course

InstID	DeptName	CourseID	SectionID	Semester	StudyYear	Title	Credits	InstName	Salary
76766	Biology	BIO-101	1	Summer	2009	Intro. to Biology	4	Crick	72000.00
76766	Biology	BIO-301	1	Summer	2010	Genetics	4	Crick	72000.00
10101	Comp. Sci.	CS-101	1	Fall	2009	Intro. to Computer Science	4	Srinivasan	65000.00
45565	Comp. Sci.	CS-101	1	Spring	2010	Intro. to Computer Science	4	Katz	75000.00
83821	Comp. Sci.	CS-190	1	Spring	2009	Game Design	4	Brandt	92000.00
83821	Comp. Sci.	CS-190	2	Spring	2009	Game Design	4	Brandt	92000.00
10101	Comp. Sci.	CS-315	1	Spring	2010	Robotics	3	Srinivasan	65000.00
45565	Comp. Sci.	CS-319	1	Spring	2010	Image Processing	3	Katz	75000.00
83821	Comp. Sci.	CS-319	2	Spring	2010	Image Processing	3	Brandt	92000.00
10101	Comp. Sci.	CS-347	1	Fall	2009	Database System Concepts	3	Srinivasan	65000.00
98345	Elec. Eng.	EE-181	1	Spring	2009	Intro. to Digital Systems	3	Kim	80000.00
12121	Finance	FIN-201	1	Spring	2010	Investment Banking	3	Wu	90000.00
32343	History	HIS-351	1	Spring	2010	World History	3	El Said	60000.00
15151	Music	MU-199	1	Spring	2010	Music Video Production	3	Mozart	40000.00
22222	Physics	PHY-101	1	Fall	2009	Physical Principles	4	Einstein	95000.00

InstID	InstName	DeptName	Salary
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califeri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

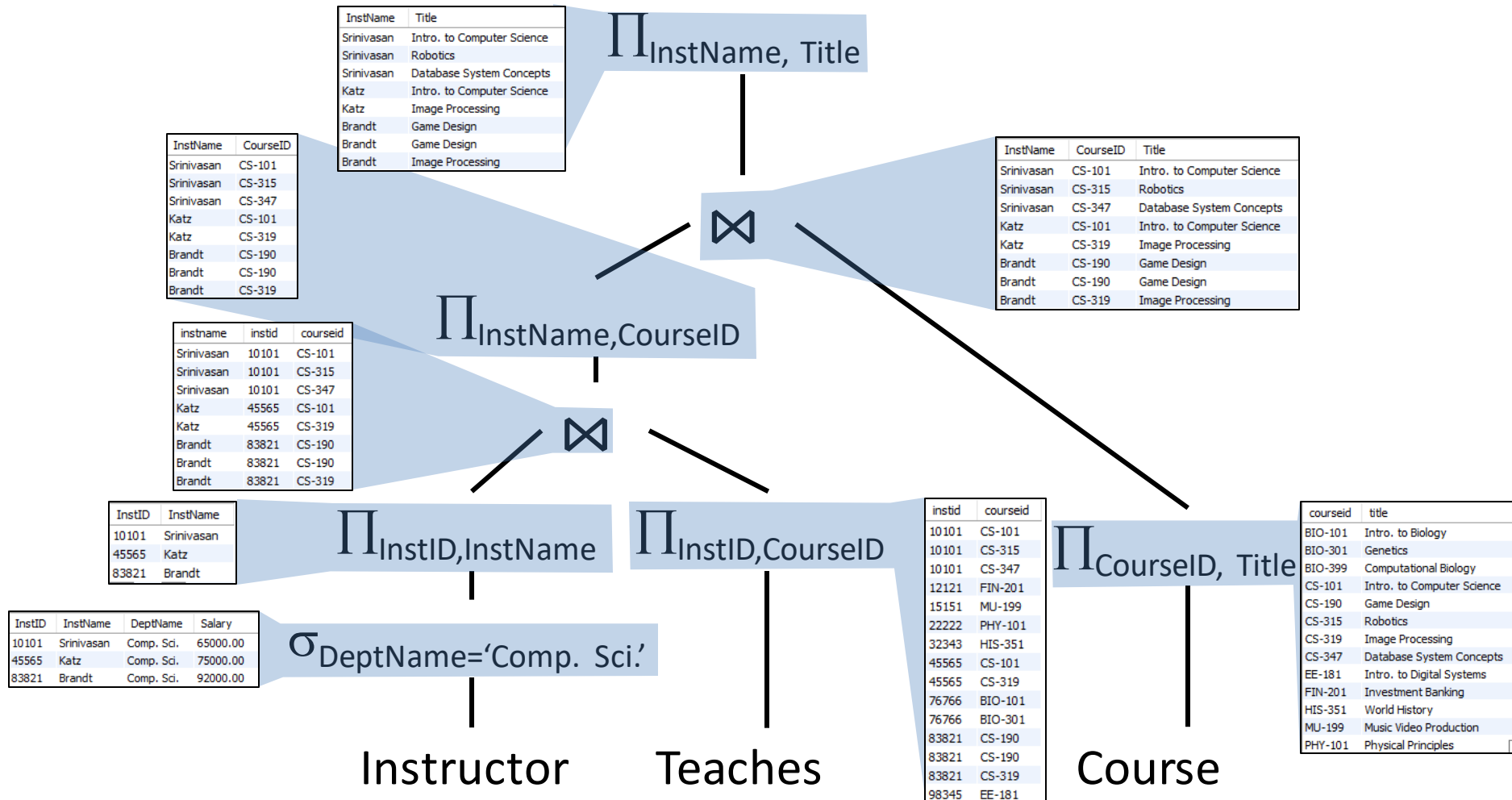
InstID	CourseID	SectionID	Semester	StudyYear
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

CourseID	InstID	SectionID	Semester	StudyYear	Title	Credits
BIO-101	76766	1	Summer	2009	Intro. to Biology	4
BIO-301	76766	1	Summer	2010	Genetics	4
CS-101	10101	1	Fall	2009	Intro. to Computer Science	4
CS-101	45565	1	Spring	2010	Intro. to Computer Science	4
CS-190	83821	1	Spring	2009	Game Design	4
CS-190	83821	2	Spring	2009	Game Design	4
CS-315	10101	1	Spring	2010	Robotics	3
CS-319	45565	1	Spring	2010	Image Processing	3
CS-319	83821	2	Spring	2010	Image Processing	3
CS-347	10101	1	Fall	2009	Database System Concepts	3
EE-181	98345	1	Spring	2009	Intro. to Digital Systems	3
FIN-201	12121	1	Spring	2010	Investment Banking	3
HIS-351	32343	1	Spring	2010	World History	3
MU-199	15151	1	Spring	2010	Music Video Production	3
PHY-101	22222	1	Fall	2009	Physical Principles	4

courseid	title	credits
BIO-101	Intro. to Biology	4
BIO-301	Genetics	4
BIO-399	Computational Biology	3
CS-101	Intro. to Computer Science	4
CS-190	Game Design	4
CS-315	Robotics	3
CS-319	Image Processing	3
CS-347	Database System Concepts	3
EE-181	Intro. to Digital Systems	3
FIN-201	Investment Banking	3
HIS-351	World History	3
MU-199	Music Video Production	3
PHY-101	Physical Principles	4

Applying optimizations

- Find the names of instructors in the Comp. Sci. department together with the course titles of the courses that the instructors teach.





Summary

- Query syntax tree
- Equivalent expressions
- Query optimizations